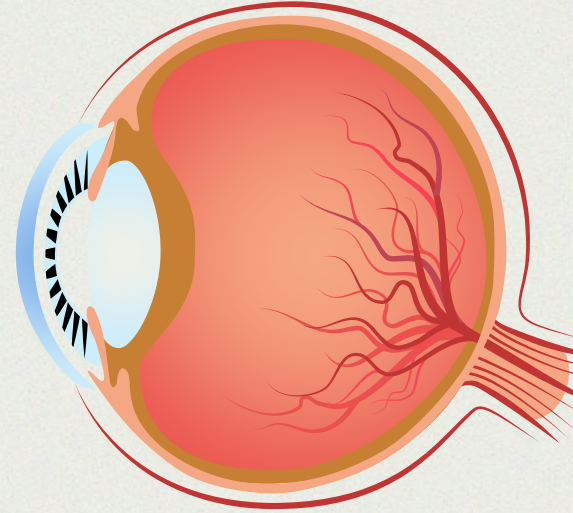


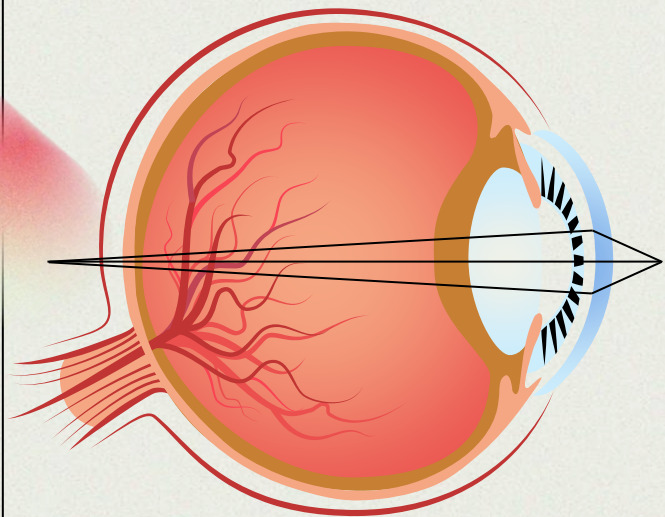
# *RETINOBLASTOMA* EYE CANCER

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Joan Gabrielle Basa

PHYS 483: Advanced Therapeutic Concepts in Oncology and Medical Physics





# 01

## THE BASICS

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# OVERVIEW<sup>[1]</sup>

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- eye cancer in the **retina**
- most common in children **younger than 5**
- average is 2 years old
- can affect 1 eye (**unilateral**) or both eyes (**bilateral**)
- can cause **blindness** and even **death** if untreated

## INCIDENCE (2019)

15

diagnosed with retinoblastoma

## MORTALITY (2022)

0

died from retinoblastoma

## FIVE-YEAR SURVIVAL

96%

in children 0 to 14 years of age

KEY STATISTICS<sup>[2]</sup><sup>[3]</sup>

# WORLDWIDE PREVALENCE (2022)<sup>[4]</sup>

**AS** ■  
2112 (52.0%)

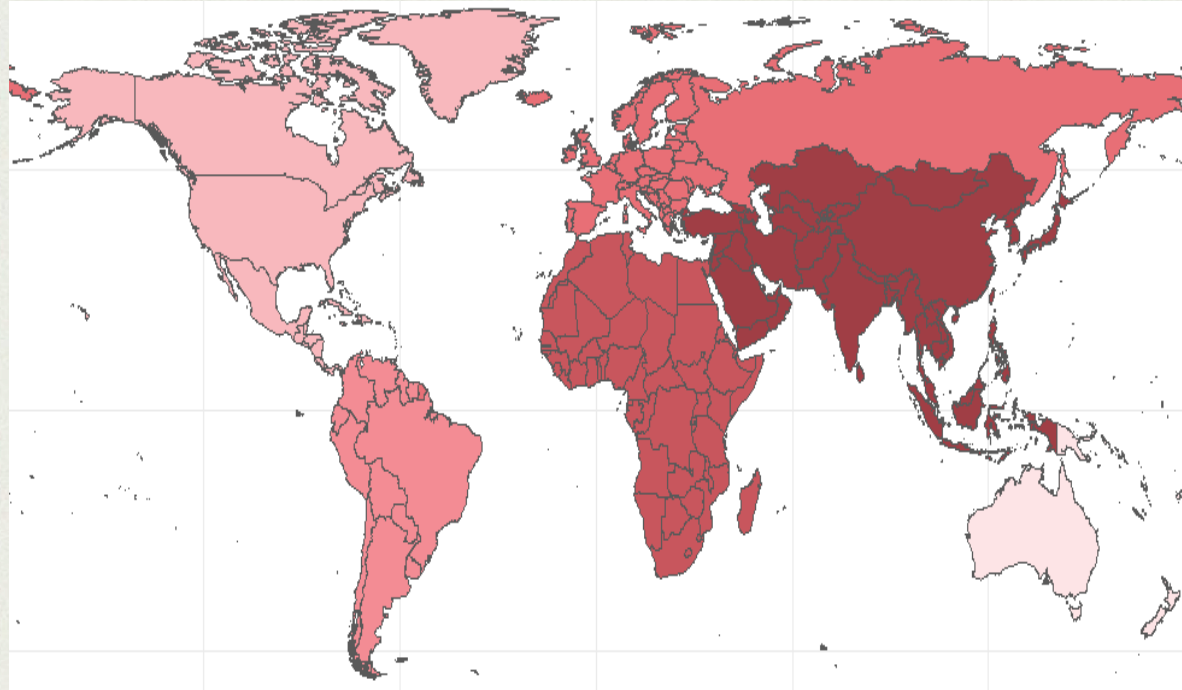
**AF** ■  
958 (23.6%)

**EU** ■  
483 (11.9%)

**SA** ■  
309 (7.6%)

**NA** ■  
182 (4.5%)

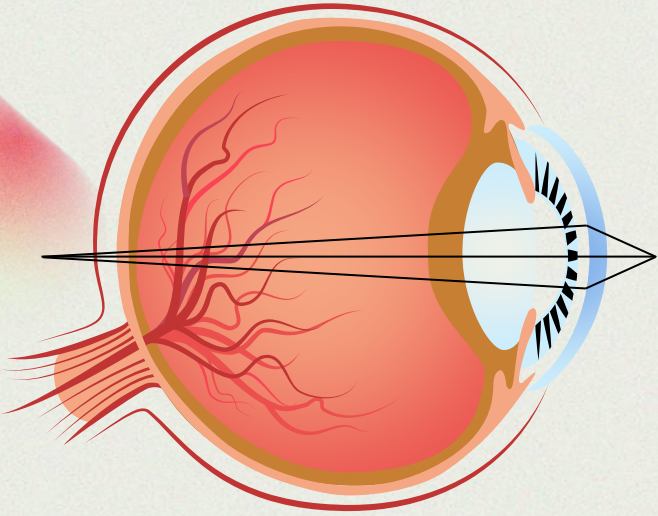
**OC** ■  
20 (0.5%)



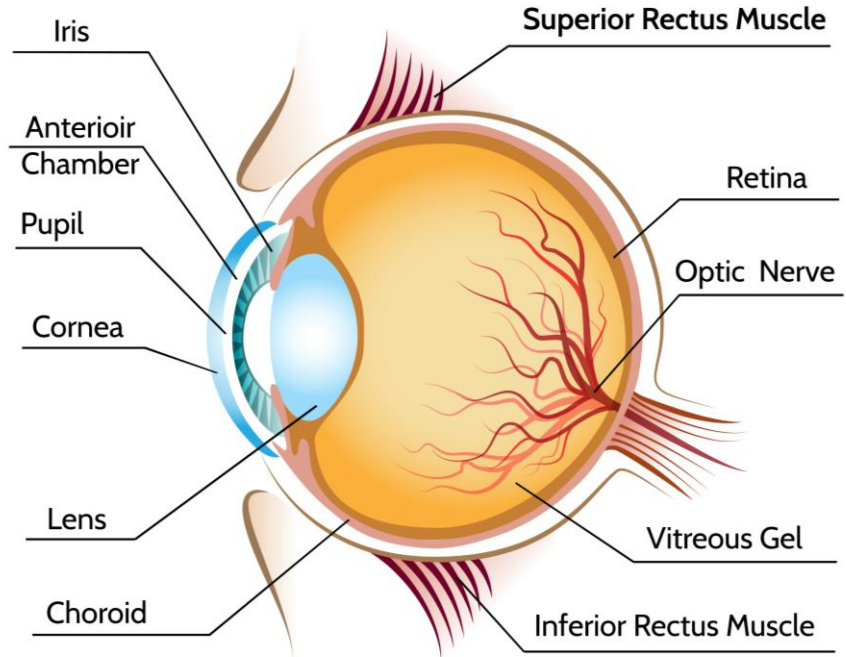
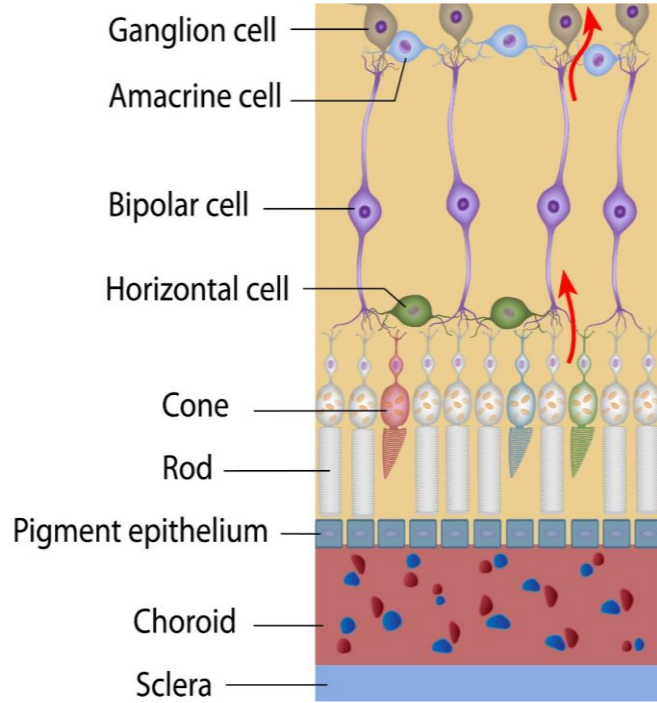
02

# PATHOPHYSIOLOGY

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# ANATOMY OF THE EYE: RETINA<sup>[5][6][7]</sup>



| FEATURES <sup>[1][5][6]</sup> | UNILATERAL                                                    | BILATERAL                                            |
|-------------------------------|---------------------------------------------------------------|------------------------------------------------------|
| Affected eyes                 | One eye                                                       | Both eyes                                            |
| Percentage                    | ~60% of cases                                                 | ~40% of cases                                        |
| Mean age at diagnosis         | 24 months                                                     | 15 months                                            |
| Tumour characteristics        | Usually unifocal (single tumour) but may be multifocal        | Usually multifocal (multiple tumours in both eyes)   |
| Progression                   | Rarely affects the second eye unless later development occurs | Both eyes are typically affected at diagnosis        |
| Family history                | Often sporadic, large tumours make it hard to confirm origin  | More likely to have a genetic or inherited component |

# GENETIC BASIS: INACTIVATION OF *RB1* GENE<sup>[6][8][9]</sup>



## HERITABLE RETINOBLASTOMA

- **Cause:** A germline mutation in one *RB1* allele is inherited, followed by a somatic mutation in a retinal cell.
- **Result:** Increased risk of bilateral and multifocal tumours.
- **Mechanism:** Loss of *RB1* function disrupts cell cycle control, leading to uncontrolled cell division and tumour formation.



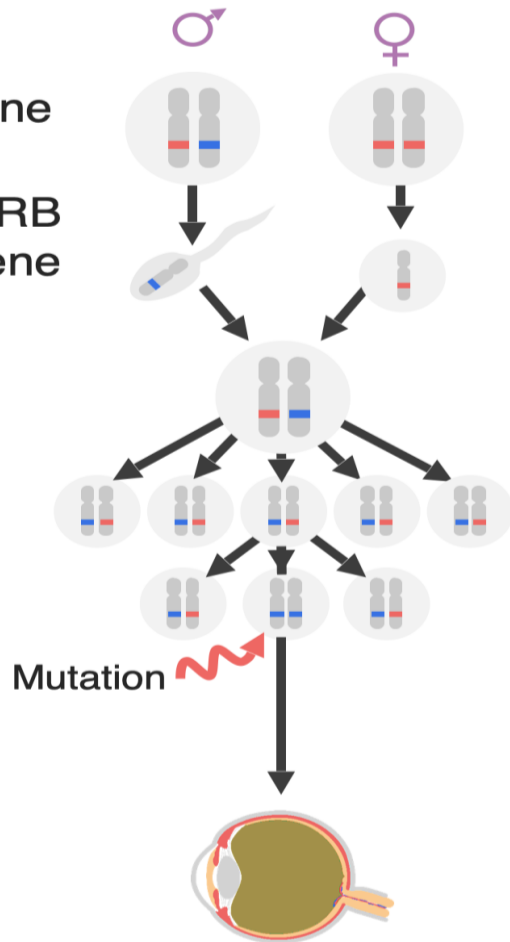
## NON-HERITABLE RETINOBLASTOMA

- **Cause:** Two somatic mutations in the *RB1* gene within a single retinal cell.
- **Result:** Typical in unilateral and unifocal tumours.
- **Mechanism:** Both *RB1* alleles are inactivated in a single retinal cell, disrupting cell cycle control.

## Heritable form

Normal gene

Mutant RB gene



Somatic cells of parents

Germ cells

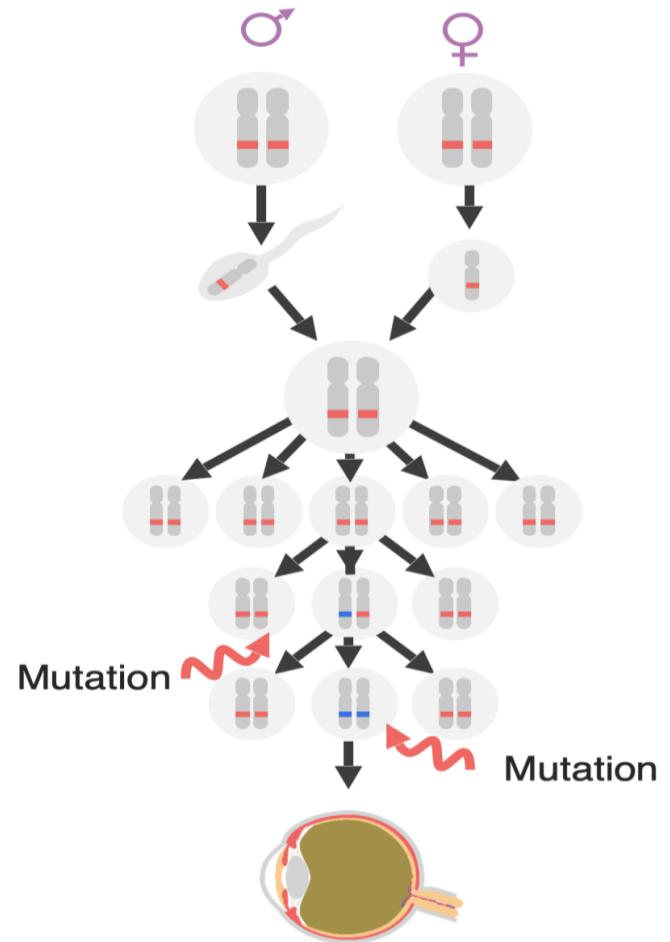
Zygote

Somatic cells of child

Retinal cells

Retinoblastoma

## Nonheritable form



# SIGNS AND SYMPTOMS<sup>[1][8]</sup>



## **01** LEUKOCORIA (WHITE PUPIL)

May notice this when taking a photo using a flash

## **02** STRABISMUS (CROSSED EYES)

Or one eye that drifts in a different direction

## **03** REDUCED VISUAL ACUITY

May fall down more often, bump into things, have trouble moving around, or hold objects close to the face

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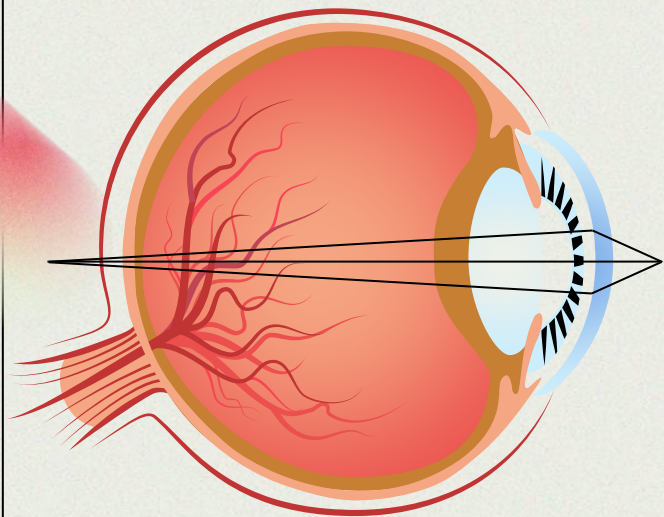
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03

# DIAGNOSIS, TREATMENT, AND MANAGEMENT

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# CLASSIFICATION → ICIR/IIRC<sup>[15]</sup>

Tumours larger than 3 mm  
in any part of the retina.

## GROUP B

Cancer has spread throughout  
the back of the eye.

## GROUP D

**A**

————

**B**

————

**C**

————

**D**

————

**E**

## GROUP A

Small tumours (3 mm or  
smaller) within the retina.

## GROUP C

Tiny areas of cancer spread  
within the back of the eye  
(seeding).

## GROUP E

No sight in the eye with  
cancer.

# DIAGNOSIS/SCREENING<sup>[8]</sup><sub>[11]</sub>



## HEALTH EXAM

child's symptoms and medical history, inc. any family history of retinoblastoma



## *RB1* GENE TEST

analyzes a sample of blood or tissue for mutations in the *RB1* gene



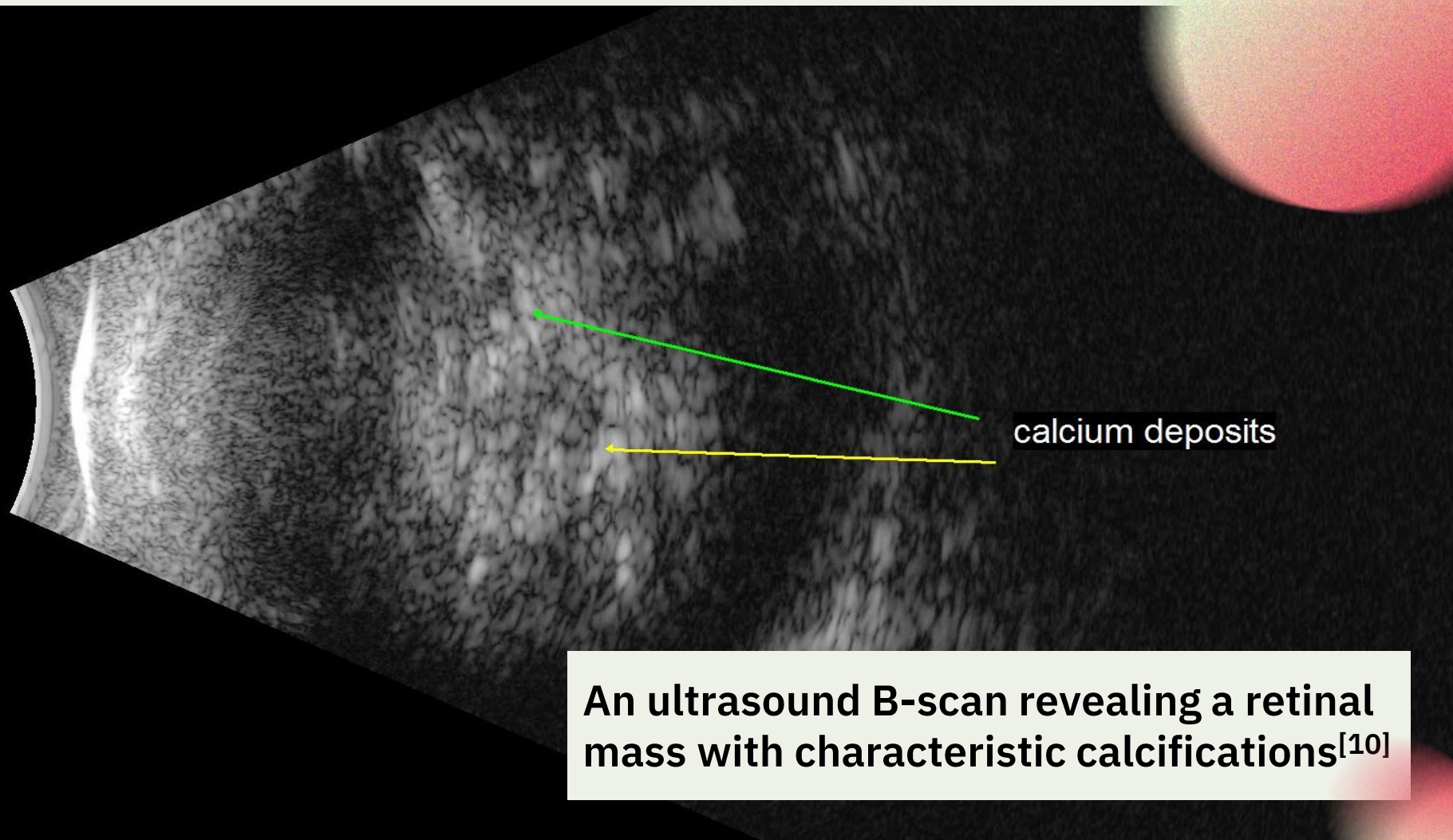
## EYE EXAM

photographic images of the inside of the eye can be taken to map any tumours



## IMAGING

to confirm the diagnosis, estimate tumour size, document intralesional calcium, and assess the spread of the tumour



calcium deposits

**An ultrasound B-scan revealing a retinal mass with characteristic calcifications<sup>[10]</sup>**

# TREATMENT<sup>[5][12][13][14]</sup>



## SURGERY

A primary treatment option that includes enucleation (or eye removal) when the tumour is too large or recurrent, or when vision cannot be salvaged.



## CHEMOTHERAPY

Utilizes drugs like carboplatin, etoposide, and vincristine to kill cancer cells, often used to shrink tumours before other treatments or to treat cancer that has spread.



## RADIOOTHERAPY

Employs high-energy rays to target and destroy cancer cells, esp. with plaque brachytherapy involving direct placement of radioactive material on the eye.

# SURGERY: ENUCLEATION<sup>[5]</sup>[12]

- Removal of the eye and part of the optic nerve
- **Reasons:** Little to no chance that vision can be saved, tumour is large and did not respond to treatment
- After enucleation, the patient will be fitted for an **artificial eye**



# CHEMOTHERAPY: INTRAVITREAL<sup>[5]</sup><sup>[13]</sup>

- Chemotherapy may be offered for intraocular retinoblastoma and may be **systemic or regional**.
- **Intravitreal chemotherapy** is a regional chemotherapy that is injected into the jelly-like substance in the eye.
- **Carboplatin + Etoposide + Vincristine**



# RADIOTHERAPY: PLAQUE BRACHYTHERAPY<sup>[5][14]</sup>

- Small pellets (**seeds**) of radioactive material are placed in a container (**a plaque or a disc**) that is placed directly on the eyeball



## BEFORE TREATMENT<sup>[8][14]</sup>

**Early diagnosis** through retinal exams and genetic counseling is crucial.

## DURING TREATMENT<sup>[8][14]</sup>

A **multidisciplinary team** tailors treatment based on tumour stage and individual needs.

## AFTER TREATMENT<sup>[8][14]</sup>

Regular **follow-up exams** monitor for recurrence and long-term effects, with continued support from specialists.

# PALLIATIVE CARE<sup>[4]</sup>

## SYMPTOM MANAGEMENT

Relieving pain  
and controlling  
other symptoms



## QUALITY OF LIFE

Ensuring the best  
possible quality  
of life



## CHILD'S WELFARE

Saving the child's  
life over saving  
their vision



## FAMILY SUPPORT

Coping with  
psychological  
stresses

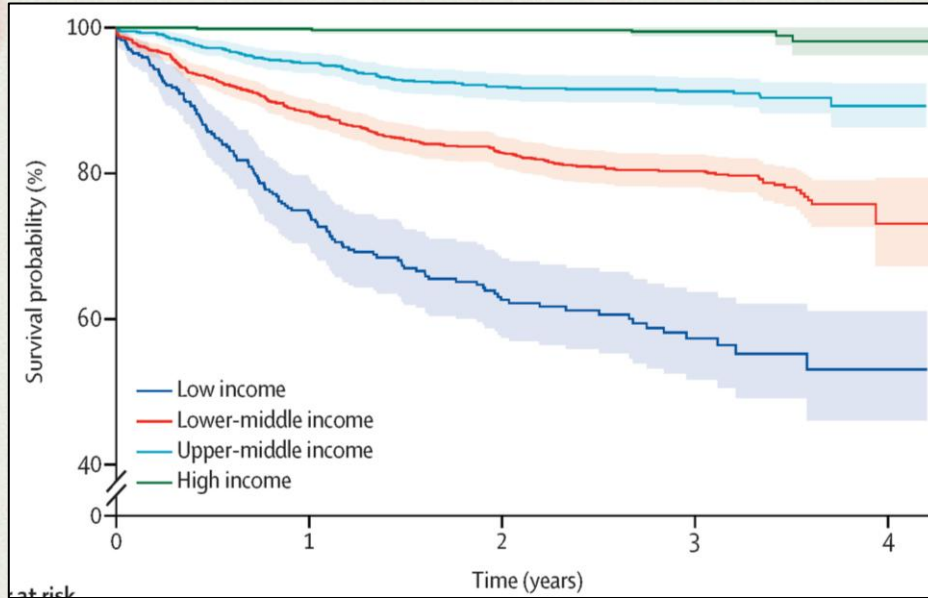


## MULTIDISCIPLINARY APPROACH

Addressing the  
various needs of  
the child and family



# PROGNOSIS AND RECURRENCE<sup>[4][9][16][17][18]</sup>



- ▶ **Group A** has *more favourable prognosis* (more likely to save vision)
- ▶ **Intraocular** has *better prognosis* than extraocular or metastatic
- ▶ Children with **trilateral retinoblastoma** have *poorer prognosis*

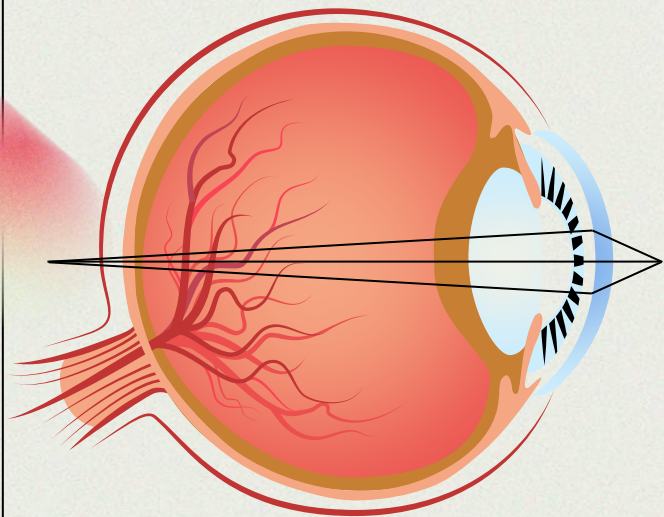
## RISK FACTORS

- Family history
- Delay in diagnosis
- Geographic location

04

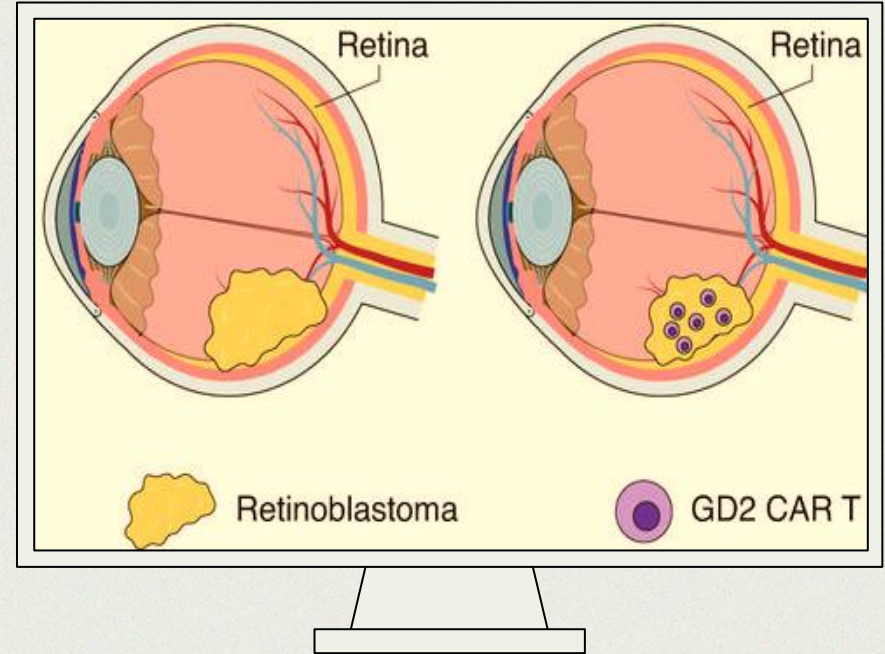
# SCIENTIFIC ADVANCES AND CHALLENGES

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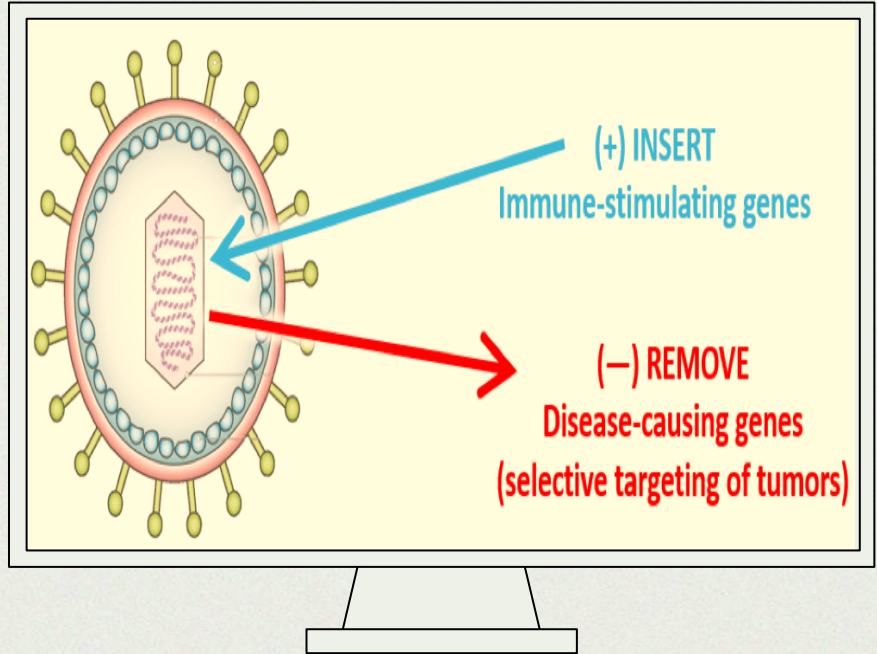
# KEY ADVANCEMENT: CAR T-Cell Therapy<sup>[19][20]</sup>

- Both **CD171** and **GD2** are effective targets on retinoblastoma cell lines.
- Targeting two different antigens by **sequential CAR T-cell applications** enhanced tumor cell killing and preempted tumour antigen loss in preclinical testing.
- Retinoblastoma cells can lose **tumour antigen expression** upon CAR T-cell encounter, with CD171 being completely lost by all tested cell lines.



# KEY ADVANCEMENT: Oncolytic Virus Therapy<sup>[21][22]</sup>

- **Virus Selection:** Scientists use a virus that can specifically target and infect retinoblastoma cells.
- **Tumour Injection:** Virus is injected directly into the tumour, where it infects and replicates inside the cancer cells.
- **Cell Destruction and Immune Activation:** The infected cancer cells burst and die, releasing substances that attack any remaining cancer cells.



# KEY CHALLENGES<sup>[23][24][25]</sup>



## 01 LIMITED AVAILABILITY

Limited tumour samples and lack of accurate models

## 02 GENETIC COMPLEXITY

Some retinoblastomas occur without *RB1* mutations

## 03 ETHICAL BARRIERS

Enrolling young patients in experimental trials requires strict safety regulations and parental consent

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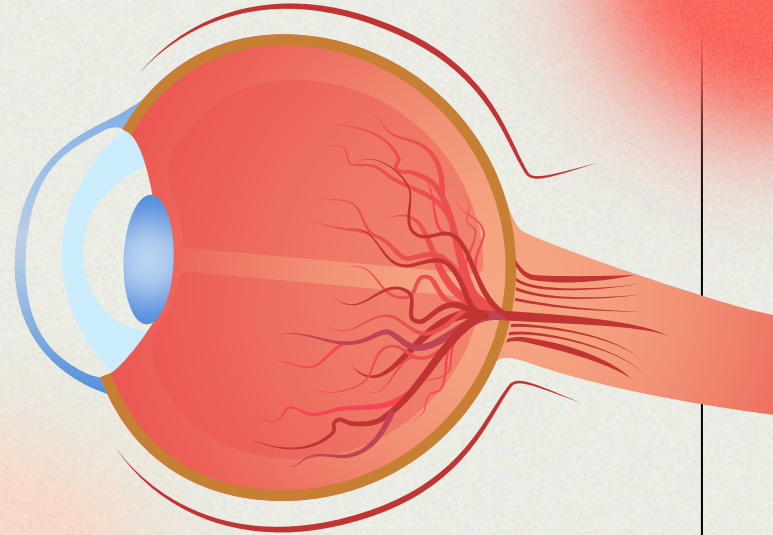
## 03 LOGISTICAL BARRIERS

Enrolling young patients in experimental trials requires strict safety regulations and parental consent



# FUTURE DIRECTIONS

- ✓ Developing **non-invasive** biopsy methods (e.g., liquid biopsy for *RB1* mutations).
- ✓ Improving **organoid and AI-based models** for drug testing.
- ✓ **Targeted molecular therapies** to reduce chemotherapy side effects.
- ✓ Global collaborations to make research findings applicable **across different populations**.



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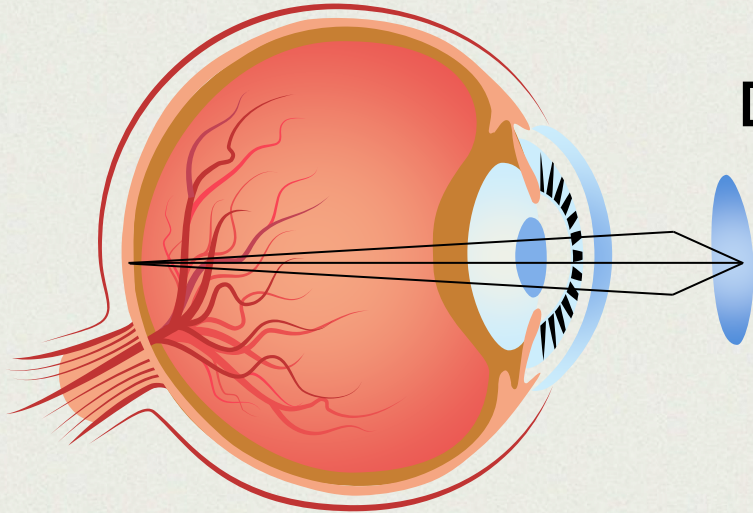
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# THANKS!

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## DO YOU HAVE ANY QUESTIONS?

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